

Abstract No.

Low-Grade Gliomas in Infancy: An Analysis of Clinical and Molecular Features

Connor Lindenmeyer¹, Svea Horn^{1,}, Amedeo M Azizi², Pablo Hernaiz Driever^{1,3}, Till Milde^{4,5}, Cornelis M Van Tillburg⁵, Philipp Sievers^{6,7}, Felix Sahm^{8,9}, Lars Behrens¹⁰, Brigitte Bison¹⁰, David Capper^{11,12}, and Arend Koch¹³*

¹Charité

²Medical University of Vienna

³German HIT-LOGGIC-Registry for pLGG in children and adolescents

⁴University Hospital Jena

⁵Hopp Children's Cancer Centre (KiTZ), Heidelberg

⁶Department of Neuropathology, Institute of Pathology, Heidelberg University Hospital, Heidelberg, Germany;

⁷German Cancer Research Center (DKFZ), Heidelberg, Germany

⁸Institute of Pathology, Heidelberg University Hospital, Heidelberg, Germany

⁹German Consortium for Translational Cancer Research (DKTK), German Cancer Research Center (DKFZ), Heidelberg, Germany

¹⁰University Hospital Augsburg

¹¹Charité-Universitätsmedizin Berlin, Corporate Member of Freie Universität Berlin, Humboldt-Universität and Berlin Institute of Health, Berlin, Germany

¹²German Cancer Consortium (DKTK), German Cancer Research Center (DKFZ), Partner Site Berlin, Berlin, Germany

¹³Charité - University Hospital Berlin

*Presenting author

Background: While pediatric low-grade gliomas (pLGG) generally exhibit favorable 10-year overall survival (OS) rates, infants comprise a high-risk cohort with increased morbidity and mortality. Utilizing molecular and clinical data from the prospective German HIT-LOGGIC registry, this study aims to detect additional prognostic risk factors within this specific patient population.

Methods: We extracted clinical, neuroradiological, histopathological, and molecular profiles from the HIT-LOGGIC registry and the LOGGIC Core Bioclinical Data Bank. Patients under two years of age with a radiologically or histologically confirmed diagnosis of pLGGs were included.

Results: In Germany, infant pLGG demonstrated an annual incidence ranging from 1.4 to 2.6 cases per 100,000 infants. Within our cohort of 177 patients, diagnosis within the first six months of life, presence of diencephalic syndrome, larger absolute tumor volume, and leptomeningeal dissemination emerged as significant independent predictors of reduced overall and progression-free survival. A markedly higher frequency of tumors had no identifiable molecular driver compared to the broader pLGG population (24% vs. 3%), alongside an enriched prevalence of BRAF V600E alterations (21% vs. 10%). Notably, the molecular profile of the eleven deceased patients did not differ compared from that of the general cohort.

Conclusions: Infants diagnosed with pLGG under six months of age experience a worse prognosis regarding survival and long-term morbidity; however, definitive molecular drivers remain elusive. Broadening the scope of advanced molecular diagnostics and investigating the role of the tumor microenvironment are critical steps for future risk stratification.



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